

Geophysical Research Letters®



RESEARCH LETTER

10.1029/2025GL117609

Key Points:

- Machine learning (ML) models show lower root mean square error (RMSE) than physics based forecasts for atmospheric river (AR) related variables over the U.S. West Coast
- Lower RMSE does not guarantee better AR detection skill
- Accurately assessing ML based weather models requires phenomenon specific and compound metrics beyond RMSE

Supporting Information:

Supporting Information may be found in the online version of this article.

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Citation:

Davis, I. W., Subramanian, A., Higgins, T. B., Sengupta, A., & Delle Monache, L. (2026). Physics-based versus AI Weather prediction models: A comparative performance assessment of Atmospheric River prediction. *Geophysical Research Letters*, 53, e2025GL117609. <https://doi.org/10.1029/2025GL117609>

Received 14 JUL 2025
Accepted 28 DEC 2025

Physics-Based Versus AI Weather Prediction Models: A Comparative Performance Assessment of Atmospheric River Prediction

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Abstract Machine learning (ML) poses a potential paradigm shift in weather forecasting, but critical questions arise regarding its ability to predict high-impact weather events. This study evaluates five state-of-the-art ML models—Aurora, GraphCast, PanguWeather, FourCastNetV2, FourCastNet—in forecasting U.S. West Coast atmospheric rivers (ARs), compared to the high-performing physics-based European Center for Medium-Range Weather Forecasts' high-resolution system (HRES) model. Analysis of 152 daily forecast cycles (November 2023–March 2024) reveals significant performance differences between the systems. While ML models often show better variable-specific root mean square error (RMSE), HRES has superior AR detection skill for the first four forecast days. PanguWeather matches HRES skill beyond day four; other ML models lag slightly. Aurora consistently exhibits the lowest AR detection performance, despite strong variable-specific RMSE metrics, highlighting a disconnect between RMSE performance and its ability to predict AR events. These findings underscore the need for phenomenon-specific metrics for ML-based numerical weather prediction model assessment and operational implementation.

Plain Language Summary Weather forecasting is undergoing a revolution with the recent emergence of machine learning (ML) models that can generate predictions in seconds, rather than hours, and show impressive accuracy on standard metrics. Unlike traditional physics-based models that simulate atmospheric processes through complex equations, these ML models learn weather patterns directly from historical data, raising important questions about their reliability for critical weather events. Atmospheric rivers—long, narrow corridors of moisture that deliver significant rainfall to the U.S. West Coast—provide an ideal test case for comparing these fundamentally different forecasting approaches. Our study evaluates five leading ML weather models against a gold-standard physics-based forecast system in predicting atmospheric rivers (ARs) on the U.S. West Coast. We discovered that, despite ML models' superior performance on standard error metrics, the traditional physics-based model more accurately detected ARs during the critical first four forecast days, when emergency decisions need to be made. Only one ML model (PanguWeather) achieved comparable performance at longer lead times. These findings suggest that conventional accuracy metrics may not accurately reflect a model's ability to predict impactful weather events, underscoring the importance of specialized evaluation methods as ML systems are integrated into operational forecasting.

1. Introduction

Machine learning (ML)-based weather prediction models have demonstrated significant advancements in accuracy and computational efficiency compared to traditional physics-based numerical weather prediction (NWP) models. Recent studies have shown that these models can achieve a root mean square error (RMSE) comparable to or better than the European Center for Medium-Range Weather Forecasts' (ECMWF) high-resolution system (HRES), while requiring orders of magnitude less computational resources. These ML-based models, including Google's GraphCast (Lam et al., 2023), NVIDIA's FourCastNet (Pathak et al., 2022) and FourCastNetV2 (Bonev et al., 2023), Huawei's PanguWeather (Bi et al., 2023), and Microsoft's Aurora (Bodnar et al., 2024), represent a paradigm shift in weather forecasting by leveraging deep learning architectures to capture complex atmospheric dynamics without explicit physical parameterizations. However, despite their impressive performance on standard metrics, these ML models exhibit important limitations that warrant closer examination. Previous work has identified specific weaknesses, including reduced sensitivity to small-scale perturbations (Selz & Craig, 2023) and systematic smoothing of sub-synoptic scale features, especially at longer lead times (Bonavita, 2024;

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Brenowitz et al., 2025; Keisler, 2022; Kochkov et al., 2024; Lam et al., 2023; Pathak et al., 2022; Rasp et al., 2024; Willard et al., 2025). These limitations raise critical questions about how ML models perform when forecasting specific high-impact weather phenomena that rely on accurately representing complex physical processes across multiple scales. One such phenomenon, atmospheric rivers (ARs), represents a particularly challenging test case for these emerging prediction systems.

Atmospheric rivers are long, narrow corridors of concentrated moisture transport in the atmosphere that can extend thousands of kilometers in length and several hundred kilometers in width (Ralph et al., 2018). These meteorological features are responsible for up to 50% of annual precipitation along the U.S. West Coast (USWC) and are associated with both beneficial water resource replenishment and devastating floods (Corringham et al., 2019; Dettinger et al., 2011; Ralph et al., 2006). The accurate prediction of AR events is critical for water resource management, flood preparation, and emergency response planning. Traditional NWP models have historically struggled with precision AR forecasting, with errors at 1-day lead times sometimes exceeding 200 km, while most affected watersheds are less than 100 km wide (Ralph et al., 2020). Furthermore, ARs provide an ideal test case for these models because their formation and evolution depend on the precise interplay of pressure systems, moisture transport, and wind fields. The ability of ML models to accurately predict ARs is therefore a crucial test of their overall forecasting capabilities and their ability to represent multiscale dynamic and thermodynamic interactions in the atmosphere. Given ARs' critical importance for regional precipitation patterns and extreme weather events, the skill of ML models in predicting ARs demands thorough investigation, yet this capability remains largely unexamined. Multiple studies have examined these models' skill in other fields, including tropical cyclone prediction (Charlton-Perez et al., 2024; DeMaria et al., 2024), extreme temperatures (Yuhang et al., 2025), and general performance (Liu et al., 2024; Rasp et al., 2024). However, this study is one of the first to examine AR prediction skill in ML weather prediction models.

As ML-based forecasting systems become increasingly integrated into operational meteorology toolboxes, it is crucial to evaluate their ability to predict high-impact weather phenomena beyond general circulation metrics. In regions where ARs drive significant hydroclimate extremes, understanding how well these emerging ML models can predict AR occurrences, intensities, and landfall characteristics is essential for both scientific advancement and public safety applications. This study aims to bridge this knowledge gap by systematically comparing the AR prediction capabilities of multiple state-of-the-art ML-based NWP models against one of the gold standards of physics-based forecasts, ECMWF's HRES model. This provides crucial insights for future operational implementation and model development.

2. Data and Methods

2.1. Forecast Models and Data

We analyze 10-day forecasts from five ML-based numerical weather prediction (NWP) models: Aurora, GraphCast, PanguWeather, FourCastNetV2, and FourCastNet alongside ECMWF's HRES as a benchmark. The forecasts used in this study are initialized daily at 00 UTC from 1 November 2023, through 31 March 2024, resulting in 152 forecast cycles per model. The ECMWF Reanalysis 5th Generation (ERA5) (Hersbach et al., 2020) serves as the ground truth for evaluating forecast accuracy. The ai-models package from the ECMWF is used to run forecasts for FourCastNet, FourCastNetV2, PanguWeather, and Aurora. The version of Aurora used is Aurora 0.25° Fine-Tuned, which is fine-tuned for HRES T0 data and showed considerably higher performance than Aurora 0.25° Pretrained (which is designed for ERA5) in preliminary testing. Through the ai-models package, only a 13-level version of GraphCast is available, so the distribution from Google DeepMind is used for the full 37-level version that is evaluated in (Lam et al., 2023). The additional levels also enable a more accurate calculation of integrated water vapor, as there is less interpolation required between the levels.

All of the AI models tested operate on a 0.25° grid, while HRES operates on a 0.1° grid. All models are regridded (after forecast quantities, such as IWV, are calculated) to a 721×1152 grid using bilinear interpolation for compatibility with the AR detection algorithm (Section 2.2). Additionally, the AI models only output 9–14 variables on 13 (all models except GraphCast) or 37 (GraphCast) pressure levels, while HRES has a vertical resolution of 137 levels, and outputs hundreds of variables. For this reason, we are unable to investigate the precipitation associated with these AR events, as most of the AI models do not output precipitation. The AI forecasts are initialized with ERA5 reanalysis, while HRES uses its real-time operational analysis. This gives the AI models a potential advantage, as reanalysis incorporates a broader data assimilation window than is available

for operational analysis. Additionally, HRES is often validated against its operational analysis, but we use ERA5 as the ground truth for all models in this study. This decision is twofold: first, ERA5 is considered the most accurate historical representation of the true atmospheric state (Hersbach et al., 2020), aligning with our goal to evaluate predictions against real-world outcomes. Second, this choice was a practical necessity, as the operational analysis for the study period was not available through open access.

2.2. Study Region, AR Detection and Attribution

Our evaluation focuses on AR activity over the USWC, defined by a bounding box of 15–65°N latitude and 170–250°E longitude. ARs are detected using the CG-Climate AR detection algorithm (Higgins et al., 2023), which is an ML based detection algorithm that uses IWV, 850 hPa winds, and sea level pressure as input fields. CG-Climate has a high level of consistency with the algorithms included in the AR Tracking Method Inter-comparison Project (ARTMIP), but has a known bias of producing spatially larger AR masks than them, consistent with its training on human-labeled examples (Higgins et al., 2023). Thus, an additional analysis using the Goldenson algorithm (Goldenson et al., 2018) at various IWV thresholds is available in the Supporting Information S1. The Goldenson algorithm is broadly similar to the all-method median of the ARTMIP algorithms on major AR metrics as reported in (Rutz et al., 2019). Forecast ARs are matched to observed ARs using a slightly modified version of the ATRISK algorithm (DeFlorio et al., 2018) with a 1,000 km distance threshold. The ATRISK algorithm consists of three steps. First, we identify AR masks using the CG-Climate algorithm. Next, we compute the IWV-weighted centroid of each AR mask, which differs from the original ATRISK method that uses integrated vapor transport-weighted centroids. Finally, we classify a forecast AR as a hit when its IWV-weighted centroid falls within 1,000 km of an AR centroid in ERA5 at the same time. For more details, see Figure 2 in (DeFlorio et al., 2018). This spatial matching approach enables quantitative evaluation of forecast accuracy in AR position and timing.

2.3. Calculation of Integrated Water Vapor (IWV)

IWV is not explicitly forecast in GraphCast, PanguWeather, or Aurora; thus for these models we must calculate it from the existing forecast fields using Equation 1,

$$IWV = \frac{1}{g} \int_{p_s}^{p_t} q dp \quad (1)$$

where g is gravitational acceleration, p_s is surface pressure, p_t is the lowest model level pressure, and q is specific humidity. 1,000 hPa is the highest pressure level of all the models, which sometimes exists either above or below the land surface or the ocean. To address this, the following approach is used:

1. *Determine surface pressure*: The hypsometric equation is applied to estimate surface pressure.
2. *Adjust for sub-surface pressure levels*:
 - If the 1,000 hPa level is below ground, q at ground level is estimated using linear interpolation between the layers immediately above and below ground.
 - If the 1,000 hPa level is above ground, the atmosphere is assumed to be well-mixed near the surface, and q at the estimated surface pressure is set equal to q at 1,000 hPa.
3. *Integrate specific humidity*: The integral in Equation 1 is performed from the calculated q and surface pressure upward to the highest model level.

2.4. Metrics

We use three standard skill metrics: Critical Success Index (CSI), Probability of Detection (POD), and False Alarm Ratio (FAR). CSI measures overall forecast accuracy for AR events, POD quantifies the fraction of observed ARs correctly predicted, and FAR indicates the proportion of ARs forecast that did not materialize. These metrics are defined as:

$$CSI = \frac{\text{Hits}}{\text{Hits} + \text{Misses} + \text{False Alarms}} \quad (2)$$

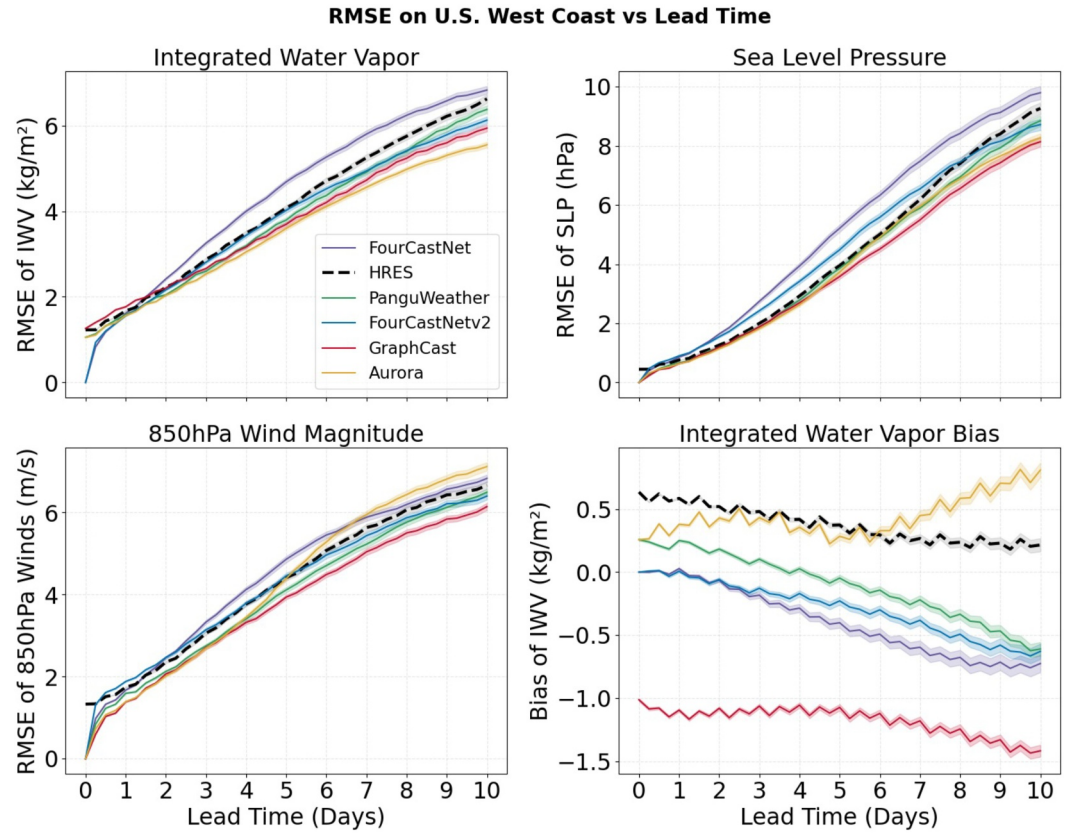


Figure 1. Root mean square error of integrated water vapor (IWV), sea level pressure, and 850 hPa winds on the USWC as a function of lead time. The bottom right panel shows bias (kg/m^2) of IWV. European Center for Medium-Range Weather Forecasts' high-resolution system is shown as a black dashed line as the benchmark, with ML-based models in various colors. Shading shows standard error.

$$\text{POD} = \frac{\text{Hits}}{\text{Hits} + \text{Misses}} \quad (3)$$

$$\text{FAR} = \frac{\text{False Alarms}}{\text{Hits} + \text{False Alarms}} \quad (4)$$

Where Hits represent ARs that were both forecast and observed, Misses are ARs that were observed but not forecast, and False Alarms are ARs that were forecast but not observed. For CSI and POD, values closer to one indicate better performance, while for FAR, values closer to 0 are better.

3. Results

Figure 1 shows the RMSE of IWV, sea level pressure, and 850 hPa winds as a function of lead time, along with IWV bias (HRES shown in black dashed line). With few exceptions, the ML-based models outperform HRES in RMSE metrics across the variables. Notable exceptions include FourCastNet (underperforming in all variables), FourCastNet-V2 (underperforming in sea level pressure at lead times 8 days or shorter), and Aurora (underperforming in 850 hPa vector winds after 5 days lead times). Interestingly, Aurora achieves the highest accuracy in forecasting individual U and V components of 850 hPa winds, but it exhibits a negative bias in both components (see Figure S1 in Supporting Information S1). This systematic underestimation leads to the reduced performance in the wind magnitude as shown in Figure 1. Improvements by ML models over HRES are most pronounced at longer lead times, closer to 10 days. This is likely due to the smoothing of these models' forecasts, which is more pronounced at longer lead times, resulting in a reduction in RMSE at the expense of a more accurate fully physical forecast state.

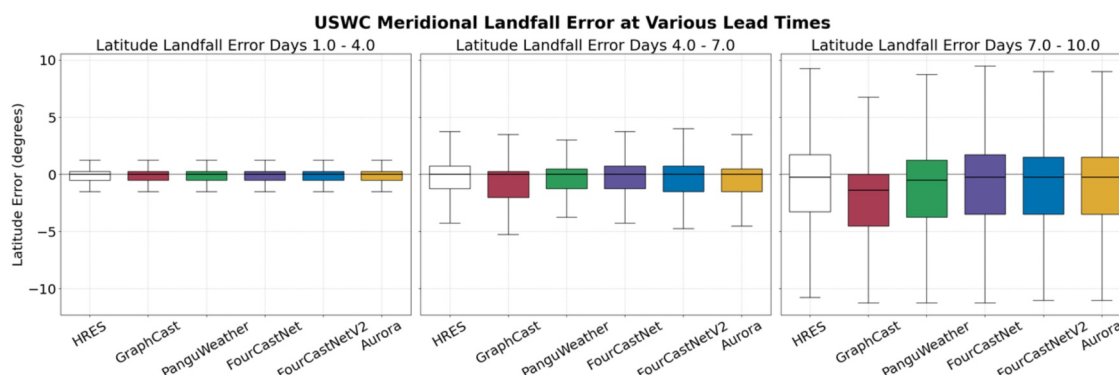


Figure 2. Box and whisker plots showing meridional landfall error of atmospheric rivers on the USWC for each model at various lead times. Boxes illustrate the 25th to 75th percentile, the centerline represents the median, and the whiskers show the extrema (disregarding outliers).

Initialization differences are evident at time 0. Only the FourCastNet models start with a zero RMSE in IWV, as they include IWV as a direct forecast field initialized with ERA5 values. In other ML models, IWV is calculated from the other forecast fields as described in Section 2.3, resulting in non-zero initial RMSE. HRES also shows non-zero RMSE at initialization across all three variables. This is due to its use of real-time analysis data as initial conditions rather than ERA5 values.

The bottom right panel of Figure 1 displays IWV bias versus lead time across all models. A systematic drying trend emerges in most models with increasing lead time, potentially indicating either moisture transport out of the USWC region or inherent limitations in moisture conservation, as physics is not explicitly enforced in any of these models, and not all conservation laws are strictly enforced in HRES. GraphCast exhibits a pronounced negative bias, beginning at approximately -1 kg/m^2 at initialization and decreasing further to -1.5 kg/m^2 by day 10. Aurora presents a distinctive pattern, initially showing slight drying before reversing to a moistening trend around day 6—the only model to display such behavior. Both FourCastNet variants demonstrate nearly identical drying patterns, starting with negligible bias at initialization before developing a negative bias approaching -0.5 kg/m^2 by day 10. PanguWeather begins with a slight positive bias (0.25 kg/m^2) that transitions to negative (-0.5 kg/m^2) by day 10. HRES maintains a consistent positive bias throughout the forecast period, starting at 0.6 kg/m^2 and declining gradually to 0.2 kg/m^2 by day 10, making it and Aurora the only models with a persistent positive bias across all lead times.

Figure 2 illustrates the meridional landfall error of ARs on the USWC across lead times. The meridional landfall error represents the difference in latitude between forecast and observed AR landfall locations, where landfall location is defined as the point on the coast with maximum IWV. During days 1–4, landfall error distributions are indistinguishable between models. By days 4–7, subtle differences emerge, with GraphCast developing a slight southerly bias in AR landfall location, though its median error remains near zero. This bias becomes more pronounced in days 7–10, with GraphCast showing a median southerly error of approximately 1.5°S . All other models, including HRES, similarly develop a slight southerly bias during this period, but much less pronounced than GraphCast, at under 1°S . Overall, except for GraphCast, all of the ML models show similar performance to HRES in meridional landfall error.

Figure 3 presents three standard skill metrics introduced in the Methods section: CSI, POD, and FAR. A consistent pattern emerges across these metrics: HRES demonstrates superior performance during the first four forecast days, after which PanguWeather achieves comparable performance, while other ML models show slightly lower skill. This performance stratification is less pronounced in FAR, where the models perform more similarly, but it is still present. At longer lead times Aurora consistently shows the lowest performance, separating from other models after day five in CSI and POD metrics, and after day eight in FAR (where lower is better). This underperformance is noteworthy given Aurora's strong RMSE performance in individual variables. This could be due to Aurora's poor representation of the 850 hPa winds, despite it achieving the lowest RMSE in the individual U and V components (Figure S1 in Supporting Information S1).

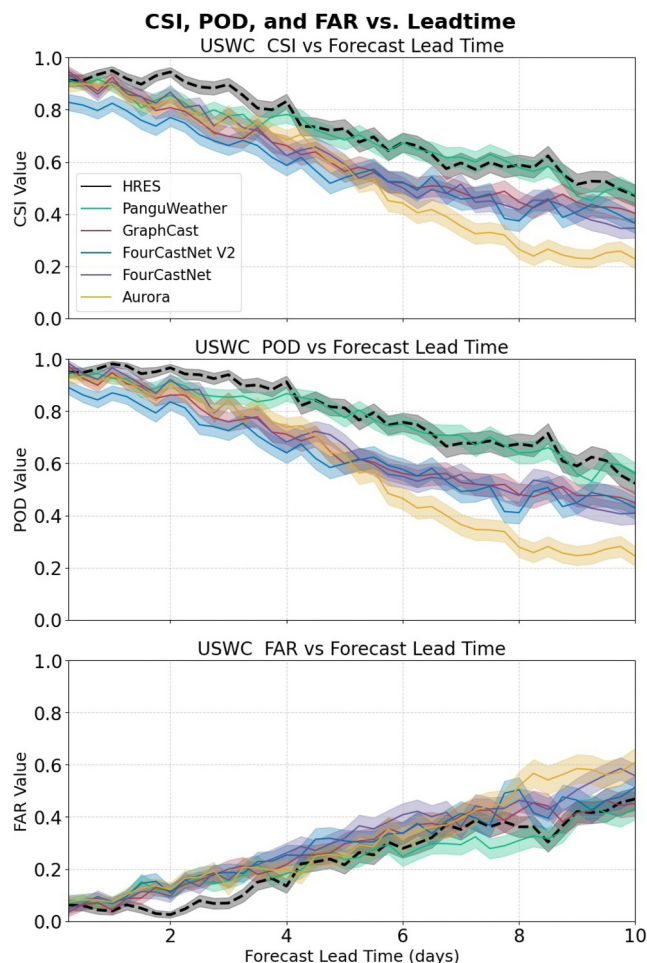


Figure 3. Critical Success Index versus lead time (top), Probability of Detection versus lead time (middle), and False Alarm Ratio versus lead time (bottom) for each model. Shading shows standard error.

Supplemental Figure S3 in Supporting Information S1 reproduces Figure 3 using the Goldenson AR detection algorithm across a range of IWV thresholds, and shows the main conclusions are broadly consistent under this alternative detection approach. We also compute IWV RMSE versus lead time conditioned on ERA5 IWV magnitude; these binned diagnostics are shown in Figure S4 in Supporting Information S1, providing a complementary evaluation based only on the physical moisture field rather than any single AR classification.

Illustrating these differences in performance is Figure 4, which presents a case study of an AR event affecting the USWC on 15 March 2024. The top panel shows the ERA5 reanalysis, while the bottom panel displays the difference between a 7 days lead time forecast from each model and the ERA5 reanalysis. Two ARs are present, but the focus will be on the eastern one making landfall on the USWC. At this lead time, Aurora has the lowest performance, while PanguWeather and HRES have the highest performance on average. We see that here, where Aurora is the only model that fails to produce the eastern AR that is impacting the USWC. In ERA5, strong southerly winds transport moisture northward along the AR core; in contrast, the Aurora forecast contains strong northerly wind anomalies in this region. This discrepancy may reflect Aurora missing a surface low pressure system at approximately 40°N, 150°W, whose cyclonic circulation drives the southerly flow in ERA5. This is more easily visible in Figure S2 in Supporting Information S1. As a result, Aurora's errors in both sea level pressure and wind fields lead to its missed AR detection in this scenario. Both HRES and PanguWeather are the highest performers in this case study, with the smallest errors in all three variables, and they also forecast the landfall location most accurately. FourCastNet forecasts the AR making landfall, but with much larger IWV, SLP, and 850 hPa wind errors. Additionally, the area of landfall is much larger than the observed AR, shifted southerly, and the shape of the AR is much different than in ERA5. FourCastNetV2 has much smaller error across the variables, but does not forecast the AR making landfall at this time. Finally, GraphCast also performs well in this case study, forecasting the AR making landfall, albeit with a southerly shift, as seen in Figure 2. Overall, despite some successes, many of the ML models have trouble representing the complexity of this individual event.

4. Conclusion

This evaluation of five ML-based NWP models against ECMWF's HRES for AR prediction reveals several important findings. While ML models generally achieve lower RMSE in basic meteorological variables (IWV, sea level pressure, and 850 hPa winds), this advantage does not consistently translate to superior AR prediction. HRES maintains superior performance in AR detection metrics (CSI, POD, and FAR) during the critical first four forecast days. Only PanguWeather achieves comparable performance to HRES beyond day four, while other ML models demonstrate slightly lower skill.

Aurora presents a particularly interesting case: despite the strongest performance in individual variable RMSE metrics, it consistently shows the lowest skill in AR detection and prediction. This disconnect highlights that improvements in traditional meteorological metrics may not necessarily translate to enhanced predictive capability for complex phenomena like ARs. The case study of 15 March 2024, further illustrates how Aurora's inability to represent the pressure field and associated wind patterns accurately leads to complete failure in AR prediction, despite its strong performance in variable-specific RMSE on average.

These findings underscore the importance of phenomenon-specific and compound metrics beyond RMSE when assessing ML-based NWP models. While these models represent a significant advancement in computational efficiency and general forecast accuracy, their application to high-impact weather phenomena requires careful

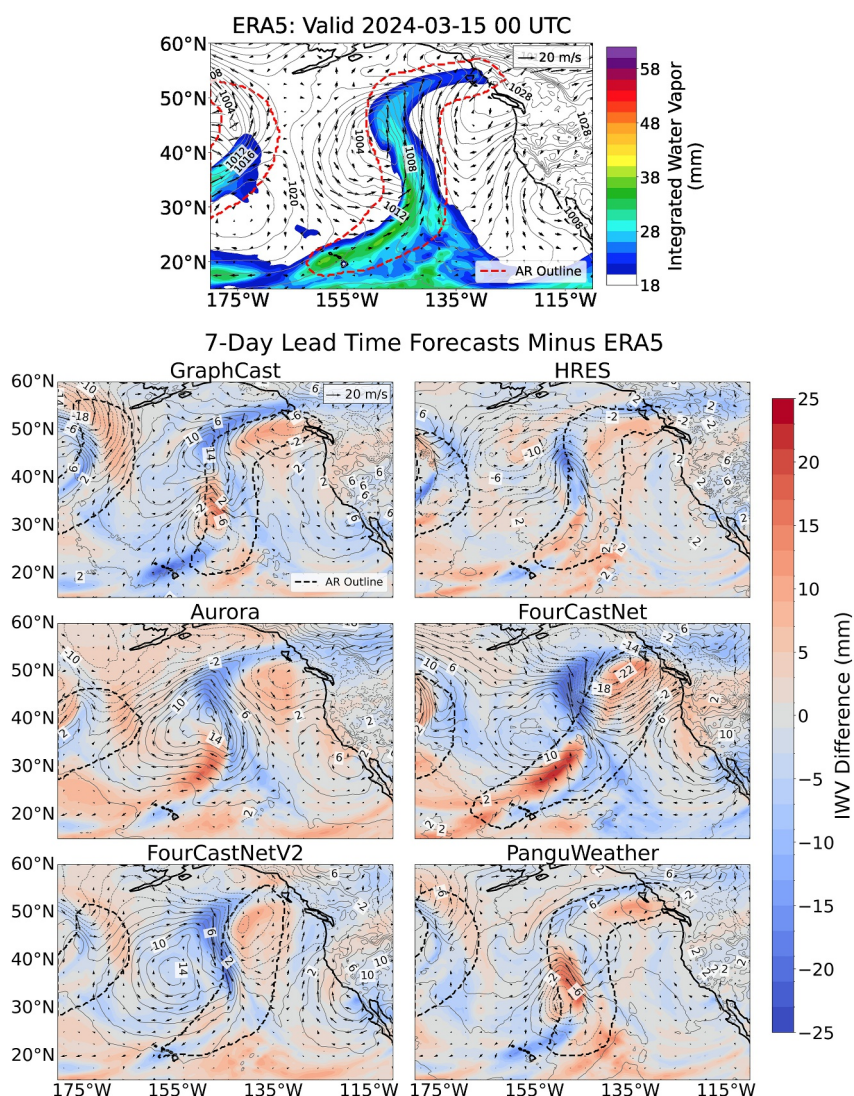


Figure 4. Case study of an atmospheric river (AR) event affecting the U.S. West Coast on 15 March 2024 (top) ERA5 reanalysis showing integrated water vapor (IWV) (mm, shading), 850 hPa winds (m/s, vectors) and sea level pressure (hPa, contours) and the AR outline in the dashed line (bottom) Difference plot showing Day 7 forecasts from all models minus ERA5 reanalysis, showing the difference IWV (mm, shading), difference in 850 hPa winds (m/s, vectors) and difference in sea level pressure (hPa, contours), and the AR outline (as detected by CG-Cliamte) in the dashed line.

consideration of their specific limitations and biases. For operational implementation, especially in regions where ARs drive significant hydroclimate extremes like the USWC, users should be aware that ML models may not yet match the performance of traditional physics-based models in all aspects of AR prediction.

However, it is important to emphasize that most of these ML models represent first-generation systems in a rapidly evolving field, and their performance is remarkably close to HRES in many aspects despite their relative novelty. PanguWeather's ability to match HRES performance beyond day four is particularly encouraging, suggesting that even these early ML implementations have significant potential. The computational efficiency of these models, generating forecasts in seconds rather than hours, points toward a promising future. This advantage is amplified when paired with an ML-based detection algorithm like CG-Cliamte, creating a computationally inexpensive end-to-end pipeline. Such a workflow opens the door for rapidly generating and analyzing huge ensembles of AR forecasts.

Future research should focus on understanding the underlying causes of these model-specific biases and limitations, which may inform refinements in ML model architecture, training approaches, or post-processing

techniques. Several critical questions remain unresolved, particularly regarding how much of the performance differences observed stem from the resolution disparities between models (HRES at 0.1° 137 levels vs. ML models at 0.25°, 13 or 37 levels). Additionally, a new class of probabilistic models is emerging, which do not produce blurred forecasts, such as GenCast (Price et al., 2025) and AIFS-CRPS (Lang et al., 2024). The performance of these models may be higher and warrants further future evaluation. Furthermore, the sensitivity of these results to the choice of AR detection algorithm warrants a more thorough investigation. Tests using a different detection method at various IWV thresholds (to emulate detection algorithms of varying permissiveness) consistently identified Aurora as the lowest performer and HRES and PanguWeather as higher performers. However, the relative rankings of the other models were not always the same as when using CG-Climate for AR detection at all lead times.

As ML NWP models continue to evolve, targeted improvements addressing their performance in predicting high-impact weather phenomena will be essential for their successful integration into operational forecasting systems. This study establishes a framework for future ML model development by pinpointing their strengths and weaknesses in AR prediction, while providing operational forecasters with evidence-based guidance for their appropriate implementation.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The minimal code and data necessary to reproduce all figures in this paper is available from Isaaciwd (2025). The raw forecasts can be freely recreated from the ai-models package (<https://github.com/ecmwf-lab/ai-models>) by the ECMWF, which was used to run forecasts for FourCastNet, FourCastNetV2, PanguWeather, and Aurora. The Google GraphCast repository (<https://github.com/google-deepmind/graphcast>) was used to generate forecasts from the GraphCast model.

Acknowledgments

This work was supported by the California Department of Water Resources Phase 3 Atmospheric River Research Program (Award 4600014294) and the Forecast Informed Reservoir Operations program (USACE Award W912HZ1920023). We also acknowledge the NSF NCAR HPC systems, managed by CISL, for providing the computational resources essential to this work. The authors also acknowledge the use of AI-powered tools, including Google's Gemini and Anthropic's Claude, to assist with code and the preparation of the manuscript.

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